

WEEK-2

Figure 1: Selecting the BookFlow database

```
Database changed
[mysql> DESCRIBE Books;
```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	
title	varchar(100)	NO		NULL	
isbn	varchar(20)	YES	UNI	NULL	
published_year	int	YES		NULL	

4 rows in set (0.002 sec)

```
[mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	Python Programming	ISBN001	2022
2	Database Systems	ISBN002	2021
3	Machine Learning	ISBN003	2024
4	Java Programming	ISBN004	2025

4 rows in set (0.000 sec)

Figure 2: Verification of the Books table structure and existing book records

```
[mysql> DESCRIBE Members;
```

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	
full_name	varchar(100)	YES		NULL	
email	varchar(100)	YES	UNI	NULL	

```
3 rows in set (0.002 sec)

[mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	Amrutha	amrutha@gmail.com
102	Rahul	rahul@gmail.com
103	Priya	priya@gmail.com

```
3 rows in set (0.000 sec)
```

Figure 3: Verification of the Members table structure

```
[mysql> DESCRIBE Loans;
```

Field	Type	Null	Key	Default	Extra
loan_id	int	NO	PRI	NULL	
member_id	int	YES	MUL	NULL	
book_id	int	YES	MUL	NULL	

```
3 rows in set (0.002 sec)

[mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id
1	101	1
2	102	2
3	103	3

```
3 rows in set (0.003 sec)
```

Figure 4: Verification of the Loans table structure and loan records

```
mysql> SELECT
  ->     m.full_name AS Member_Name,
  ->     b.title AS Book_Title
  -> FROM Loans l
  -> INNER JOIN Members m
  ->     ON l.member_id = m.member_id
  -> INNER JOIN Books b
  ->     ON l.book_id = b.book_id;
```

Member_Name	Book_Title
Amrutha	Python Programming
Rahul	Database Systems
Priya	Machine Learning

```
3 rows in set (0.001 sec)
```

Figure 5: INNER JOIN displaying member names and the books borrowed

```
mysql> SELECT
    ->     published_year,
    ->     COUNT(book_id) AS Total_Books
    -> FROM Books
    -> GROUP BY published_year
    -> ORDER BY published_year;
```

published_year	Total_Books
2021	1
2022	1
2024	1
2025	1

4 rows in set (0.003 sec)

Figure 6: Grouping books by publication year and counting the total number of books

```
[mysql> DESCRIBE Donation_History;
```

Field	Type	Null	Key	Default	Extra
donation_id	int	NO	PRI	NULL	
book_id	int	YES	MUL	NULL	
category	varchar(50)	YES		NULL	
donation_date	date	YES		NULL	

4 rows in set (0.002 sec)

```
[mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	category	donation_date
1	4	Programming	2026-08-03

1 row in set (0.003 sec)

Figure 7: Verification of the Donation_History table structure and existing donation records

```
mysql> CREATE INDEX idx_books_isbn
```

```
    -> ON Books(isbn);
```

Query OK, 0 rows affected, 1 warning (0.037 sec)

Records: 0 Duplicates: 0 Warnings: 1

Figure 8: Creating an index on the ISBN column for faster book searches

```
mysql> SELECT *
      -> FROM Books
      -> WHERE isbn = 'ISBN003';
```

book_id	title	isbn	published_year
3	Machine Learning	ISBN003	2024

```
1 row in set (0.002 sec)
```

Figure 9: Searching for a book using its indexed ISBN

```
[mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	Python Programming	ISBN001	2022
2	Database Systems	ISBN002	2021
3	Machine Learning	ISBN003	2024
4	Java Programming	ISBN004	2025

```
4 rows in set (0.035 sec)

mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	Amrutha	amrutha@gmail.com
102	Rahul	rahul@gmail.com
103	Priya	priya@gmail.com

```
3 rows in set (0.001 sec)

mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id
1	101	1
2	102	2
3	103	3

```
3 rows in set (0.018 sec)

mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	category	donation_date
1	4	Programming	2026-08-03

```
1 row in set (0.003 sec)
```

Figure 10: Final Books table ,Members table ,Loans table and Donation_History table